

WEEK-1

OUT-PUT EXERCISE-4

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:E:\IntelliJ2026\IntelliJ IDEA 2026.1.2\lib\idea_rt.jar=5530
Employee Added
Employee Added
Employee Added

Employees:
101 Rahul Manager 70000.0
102 Priya Developer 60000.0
103 Kiran Tester 50000.0

Search Employee:
102 Priya Developer 60000.0
Employee Deleted

After Deletion:
101 Rahul Manager 70000.0
103 Kiran Tester 50000.0

Process finished with exit code 0
```

Explanation:

Array Representation in Memory

An array is a collection of elements stored in **contiguous memory locations**. Each element occupies a fixed amount of memory, allowing direct access using its index.

Time Complexity

Operation	Time Complexity
Add Employee	O(1)
Search Employee	O(n)
Traverse Employees	O(n)
Delete Employee	O(n)

Explanation

- **Add:** Inserts at the next available position.
- **Search:** May need to check all employees.
- **Traverse:** Visits every employee exactly once.
- **Delete:** Requires shifting elements after deletion.

Limitations of Arrays

1. Fixed Size

- The size must be decided beforehand.
- Cannot grow dynamically.

2. Costly Deletion

- Elements must be shifted after deletion.

3. Costly Insertion

- Inserting in the middle requires shifting elements.

4. Memory Wastage

- Unused allocated space remains occupied.